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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

B.Tech. AI-DS (C Section)

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TECHNICAL REPORT

ON

WATER QUALITY MONITORING SYSTEM

AN ARDUINO-BASED pH AND TDS MONITORING SYSTEM WITH LCD DISPLAY AND LED / BUZZER ALERTS

SUBJECT NAME : IoT AND SENSOR TECHNOLOGIES

SUBJECT CODE : 24AIPC501

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1. Abstract

Access to safe drinking water is a basic requirement for public health, yet many households, schools and small communities have no quick way to check whether the water they use is fit for consumption. Laboratory testing is accurate but slow and expensive, and it is rarely available on demand.

This project presents a low-cost, portable **Water Quality Monitoring System** built around an Arduino microcontroller. The system measures two key indicators of water quality: **pH**, which shows how acidic or alkaline the water is, and **TDS (Total Dissolved Solids)**, which shows the concentration of dissolved minerals and salts in parts per million (ppm). The readings are displayed continuously on a 16x2 LCD.

The measured values are compared with the limits recommended for drinking water. When the water is within limits, a green LED glows. When either parameter is outside its safe range, a red LED turns on, a buzzer sounds, and the LCD identifies the parameter that failed. The design is simple, inexpensive and easy to assemble, and it can be extended with additional sensors or wireless connectivity.

Keywords: water quality, pH sensor, TDS sensor, Arduino, LCD, LED, buzzer, embedded system.

2. Introduction and Problem Statement

2.1 Background

Water is used for drinking, cooking, agriculture and industry, and its quality directly affects health. Contamination can come from industrial discharge, agricultural runoff, aging pipes, sewage leakage and natural minerals in the ground. Poor water may look clear and still be unsafe, so visual inspection alone cannot be trusted.

Two simple electrical measurements give a useful first indication of quality. pH reveals chemical balance, and TDS reveals the amount of dissolved material. Neither parameter can prove that water is completely safe, but abnormal values are a reliable warning that further testing or treatment is needed.

2.2 Problem Statement

Existing laboratory methods are costly and slow, while commercial handheld meters give a number but no clear safe or unsafe decision and no alarm. There is a need for an affordable device that measures water quality continuously, interprets the result automatically, and alerts the user immediately.

2.3 Objectives

- To measure the pH of a water sample using a pH sensor.
- To measure TDS in ppm using a TDS sensor, with optional temperature compensation.
- To display both values in real time on a 16x2 LCD.
- To indicate safe or unsafe water using a green and red LED.
- To raise an audible alarm with a buzzer when any limit is exceeded.
- To keep the system low-cost, portable and easy to build and maintain.

2.4 Scope

The project covers on-site, quick screening of drinking and domestic water. It does not replace laboratory analysis and does not detect bacteria, heavy metals, pesticides or other specific chemical contaminants. Its purpose is early warning and awareness.

3. Water Quality Parameters

3.1 pH

pH is a measure of the hydrogen ion concentration in water on a scale from 0 to 14. A value of 7 is neutral, values below 7 are acidic and values above 7 are alkaline. Drinking water is normally expected to lie between **6.5 and 8.5**. Very low pH can corrode pipes and release metals such as lead and copper, while very high pH gives a bitter taste, scaling and reduced disinfection efficiency.

3.2 Total Dissolved Solids (TDS)

TDS is the total amount of dissolved inorganic salts and small amounts of organic matter in water, expressed in ppm or mg/L. Typical dissolved substances include calcium, magnesium, sodium, potassium, chlorides, sulphates and bicarbonates. A TDS meter does not count the particles directly. It measures electrical conductivity (EC), because dissolved ions conduct current, and then converts it to ppm using a factor (commonly 0.5).

TDS (ppm) = EC (µS/cm) x 0.5

3.3 Interpretation Tables

TDS (ppm)	Water quality	Remark
0 - 50	Very low mineral content	Distilled / RO water; may taste flat
50 - 150	Excellent	Good for drinking
150 - 300	Good	Acceptable for drinking
300 - 500	Fair	Acceptable limit for drinking water
Above 500	Poor / unsafe	Not recommended; treatment advised

pH value	Nature	Effect
Below 6.5	Acidic	Corrosion, metallic taste
6.5 - 8.5	Safe range	Suitable for drinking
Above 8.5	Alkaline	Bitter taste, scaling, poor disinfection

Limits follow commonly used WHO and BIS guidance for drinking water. Check the standard that applies in your region before relying on them.

4. Hardware Components

The system uses a small number of readily available modules. All of them operate from a 5 V supply, which keeps the design simple and safe.

Component	Specification	Function
Arduino Uno / Nano	ATmega328P, 10-bit ADC, 5 V logic	Reads sensors, performs calculations and controls outputs
pH sensor module	Analog probe with BNC and signal board, 0-14 pH	Produces a voltage proportional to hydrogen ion activity
TDS sensor module	Analog, 0-1000 ppm, 3.3-5.5 V	Measures conductivity of water and outputs an analog voltage
16x2 LCD with I2C	HD44780 with PCF8574 backpack, address 0x27	Displays pH, TDS and status text using only two data wires
LED (green)	5 mm, with 220 ohm resistor	Indicates that water is within safe limits
LED (red)	5 mm, with 220 ohm resistor	Indicates unsafe water
Buzzer	Active or passive piezo, 5 V	Gives an audible alarm for unsafe water
Resistors	2 x 220 ohm	Limit LED current
Breadboard / PCB, wires	Standard	Interconnection
Power supply	5 V USB or 9 V adapter	Powers the whole system

4.1 Role of Each Block

- **Sensors:** convert the physical properties of water into analog voltages.
- **Microcontroller:** digitises the voltages, converts them to engineering units and applies the decision logic.
- **Display:** gives the user a continuous numerical readout.
- **LED and buzzer:** provide an immediate visual and audible decision, useful even when nobody is reading the display.

4.2 Selection Notes

An Arduino Uno was chosen for its large community, simple programming and built-in ADC. The I2C LCD saves pins, leaving room for future sensors. Analog sensor modules with onboard signal conditioning avoid the need for external amplifier circuits. The probes are the most delicate parts and should be kept clean, stored as the manufacturer advises, and not left dry for long periods.

5. Circuit Design and Connections

5.1 System Block Diagram

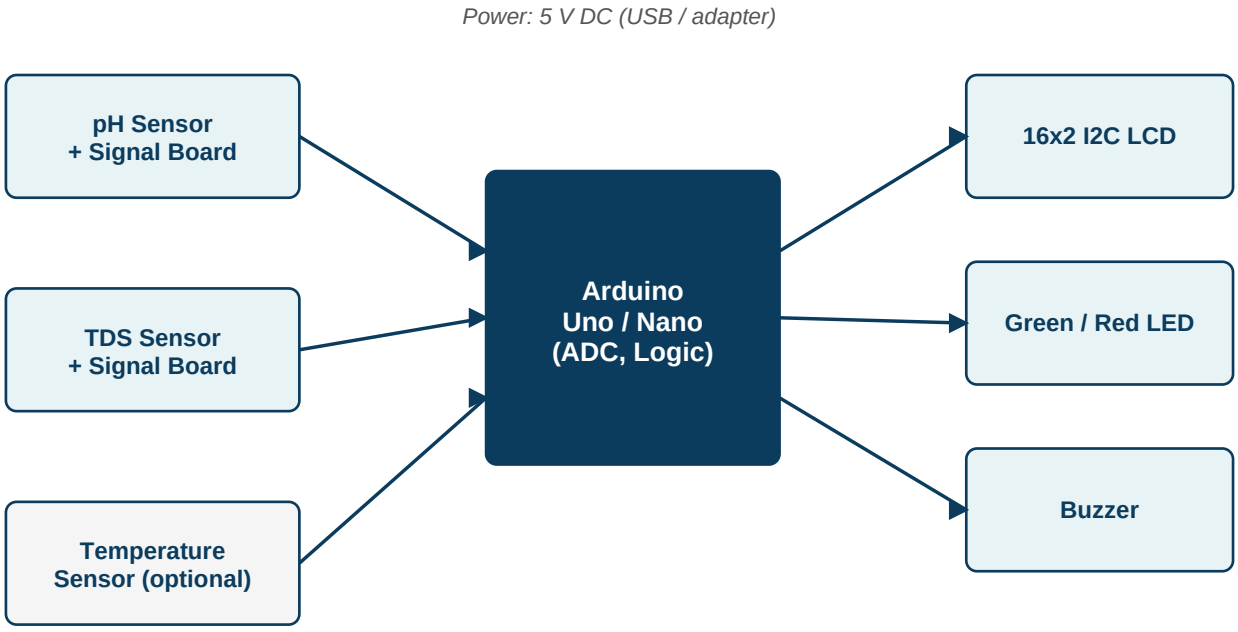


Figure 1: Block diagram of the water quality monitoring system

5.2 Pin Connections

Device	Device pin	Arduino pin
pH sensor board	Po (analog out)	A0
pH sensor board	VCC / GND	5V / GND
TDS sensor board	A (analog out)	A1
TDS sensor board	VCC / GND	5V / GND
LCD (I2C)	SDA / SCL	A4 / A5
LCD (I2C)	VCC / GND	5V / GND
Green LED	Anode via 220 ohm	D6
Red LED	Anode via 220 ohm	D7
Buzzer	Positive	D8
LEDs and buzzer	Cathode / negative	GND

Note: Pins A4 and A5 are the hardware I2C pins on the Uno and Nano. If the LCD stays blank, adjust its contrast potentiometer and verify the I2C address (0x27 or 0x3F) with an I2C scanner sketch.

6. Working Principle and Calculations

6.1 Operation

The probes are immersed in the water sample. Each sensor module outputs an analog voltage between 0 and 5 V. The Arduino's 10-bit ADC converts this voltage to a number from 0 to 1023. To reduce noise, the program takes 30 samples and averages them before every calculation. The averaged ADC value is converted back to a voltage, then to pH and TDS using the formulas below.

6.2 ADC to Voltage

Voltage (V) = ADC value x 5.0 / 1024

For example, an ADC reading of 512 corresponds to 512 x 5.0 / 1024 = 2.50 V.

6.3 pH Calculation

A pH electrode produces a voltage that changes almost linearly with pH. Two buffer solutions (pH 7.00 and pH 4.01) are used to calibrate it. The slope is found from the two measured voltages and the pH is obtained from a straight-line equation.

Slope = (7.00 - 4.01) / (V7 - V4)
pH = 7.00 + Slope x (V - V7)

Example: with V7 = 2.50 V and V4 = 3.05 V, the slope is 2.99 / (-0.55) = -5.44 pH per volt. A measured voltage of 2.40 V then gives pH = 7.00 + (-5.44)(2.40 - 2.50) = 7.54.

6.4 TDS Calculation with Temperature Compensation

Conductivity rises with temperature by roughly 2 percent per degree Celsius, so the voltage is first corrected to the 25 degree C reference:

Compensation coefficient = 1.0 + 0.02 x (T - 25)
Compensated voltage = V / coefficient

The compensated voltage is converted to ppm by the third-order polynomial supplied with the common analog TDS module:

TDS (ppm) = (133.42 V^3 - 255.86 V^2 + 857.39 V) x 0.5

6.5 Decision Logic

Condition	LED	Buzzer	LCD status
pH 6.5-8.5 and TDS below 500 ppm	Green ON	OFF	OK / OK
pH outside 6.5-8.5	Red ON	ON	pH BAD
TDS above 500 ppm	Red ON	ON	TDS BAD
Both out of range	Red ON	ON	pH BAD / TDS BAD

7. Software Algorithm and Flowchart

The firmware is written in the Arduino C++ environment and uses the LiquidCrystal_I2C library for the display. The program has two parts: a one-time setup, and a loop that repeats forever.

7.1 Algorithm

- Initialise the LCD, LED pins and buzzer pin; show a start-up message.
- Read the pH channel 30 times and average the result.
- Read the TDS channel 30 times and average the result.
- Convert both voltages to pH and TDS (ppm); apply temperature compensation to TDS.
- Print the values on the LCD.
- Compare the values with the limits. If both are inside, turn on the green LED and keep the buzzer off.
- If either is outside, turn on the red LED, sound the buzzer and flag the parameter on the LCD.
- Wait one second and repeat.

7.2 Flowchart

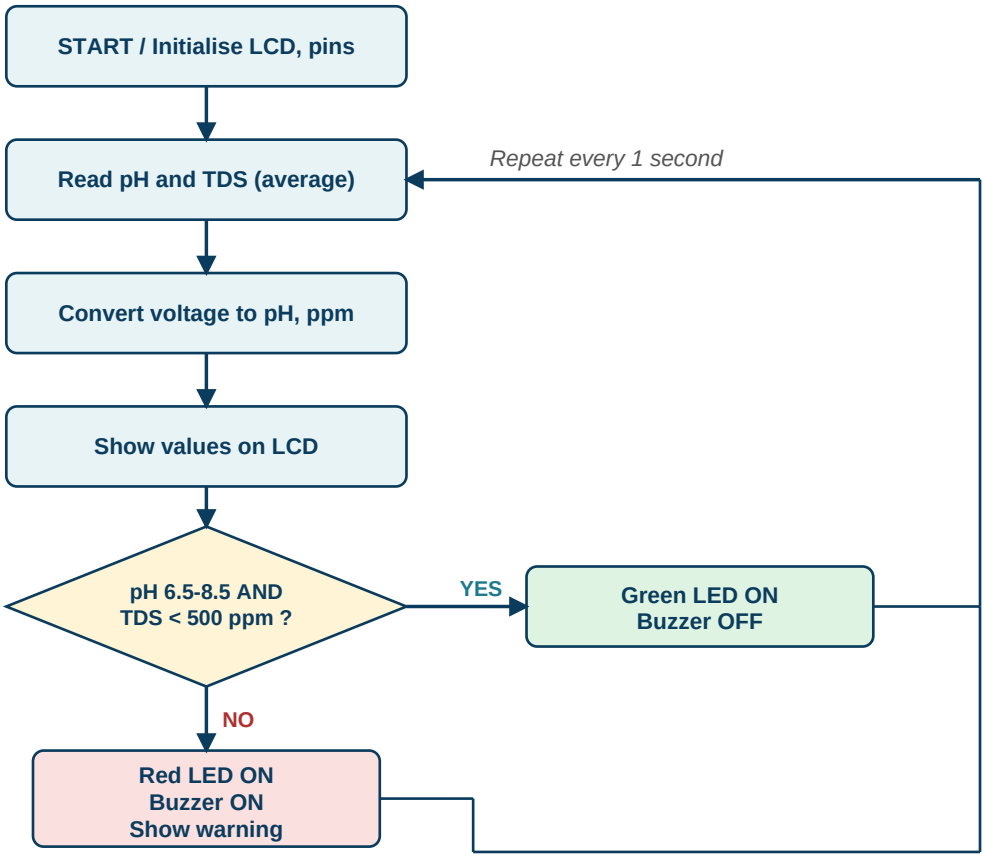


Figure 2: Program flowchart

8. Arduino Source Code

Replace the calibration voltages with your own measurements before use.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

#define PH_PIN    A0
#define TDS_PIN   A1
#define GREEN_LED 6
#define RED_LED   7
#define BUZZER    8

const float VREF = 5.0;            // ADC reference voltage
const int   SAMPLES = 30;          // readings to average
const float PH_LOW = 6.5, PH_HIGH = 8.5;
const float TDS_LIMIT = 500.0;     // ppm
float temperature = 25.0;          // deg C (fixed or from sensor)

// pH calibration (two-point): measure V at pH 7.00 and pH 4.01
float V_PH7 = 2.50;   // volts in pH 7 buffer  (replace)
float V_PH4 = 3.05;   // volts in pH 4 buffer  (replace)

float readAverage(int pin) {
  long sum = 0;
  for (int i = 0; i < SAMPLES; i++) { sum += analogRead(pin); delay(10); }
  return (sum / (float)SAMPLES) * VREF / 1024.0;   // volts
}

float voltageToPH(float v) {
  float slope = (7.0 - 4.01) / (V_PH7 - V_PH4);
  return 7.0 + slope * (v - V_PH7);
}

float voltageToTDS(float v) {
  float comp = 1.0 + 0.02 * (temperature - 25.0);
  float cv = v / comp;
  return (133.42*cv*cv*cv - 255.86*cv*cv + 857.39*cv) * 0.5;
}

void setup() {
  pinMode(GREEN_LED, OUTPUT); pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);
  lcd.init(); lcd.backlight();
  lcd.setCursor(0,0); lcd.print("Water Quality");
  lcd.setCursor(0,1); lcd.print("Monitor Ready");
  delay(2000); lcd.clear();
}

void loop() {
  float ph  = voltageToPH(readAverage(PH_PIN));
  float tds = voltageToTDS(readAverage(TDS_PIN));
  if (tds < 0) tds = 0;

  bool phBad  = (ph < PH_LOW || ph > PH_HIGH);
  bool tdsBad = (tds > TDS_LIMIT);

  lcd.setCursor(0,0);
  lcd.print("pH:"); lcd.print(ph,1); lcd.print("  ");
  lcd.print(phBad ? "BAD " : "OK  ");
  lcd.setCursor(0,1);
  lcd.print("TDS:"); lcd.print((int)tds); lcd.print("ppm ");
  lcd.print(tdsBad ? "BAD" : "OK ");

  if (phBad || tdsBad) {
    digitalWrite(GREEN_LED, LOW);  digitalWrite(RED_LED, HIGH);
    tone(BUZZER, 2000, 300);
  } else {
    digitalWrite(GREEN_LED, HIGH); digitalWrite(RED_LED, LOW);
    noTone(BUZZER);
  }
  delay(1000);
}
```

9. Testing, Calibration and Results

9.1 Calibration Procedure

- Rinse the pH probe with clean water and place it in pH 7.00 buffer. Wait for the voltage to settle and record it as V7.
- Rinse again and place the probe in pH 4.01 buffer. Record the voltage as V4.
- Enter V7 and V4 in the code and upload it. Check a third buffer (for example pH 9.18) to confirm accuracy.
- Verify the TDS sensor with a standard solution (for example 342 ppm NaCl) and adjust the conversion factor if needed.
- Repeat calibration every few weeks or when readings drift.

9.2 Sample Test Readings

The values below are illustrative sample readings that show the expected format of the results table. Replace them with your own measurements from the built prototype.

Sample	pH	TDS (ppm)	Status	LED / Buzzer
Distilled water	6.9	12	Safe (very low minerals)	Green / Off
Tap water	7.4	185	Safe	Green / Off
RO purified water	7.1	42	Safe	Green / Off
Borewell water	7.8	450	Safe (near limit)	Green / Off
Hard water	8.1	620	Unsafe (TDS)	Red / On
Lemon water	3.0	90	Unsafe (pH)	Red / On

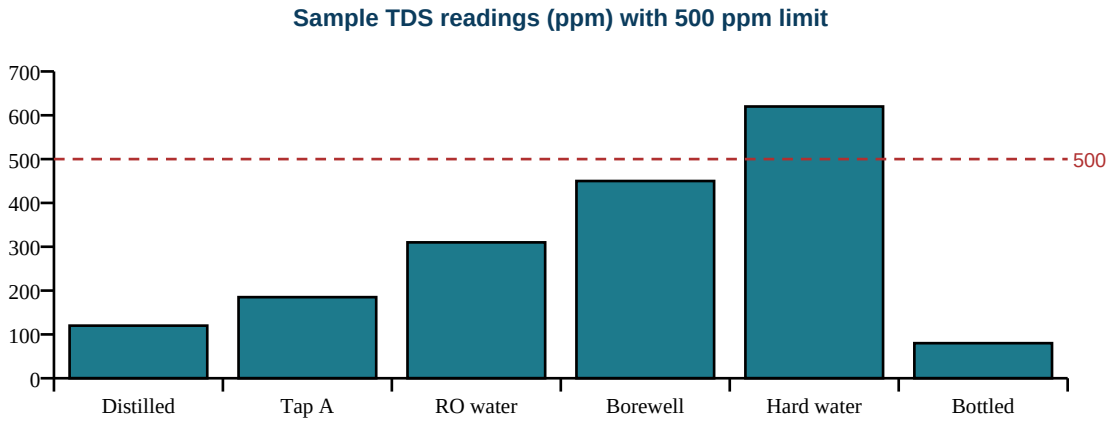


Figure 3: Sample TDS readings against the 500 ppm limit

9.3 Observations

The system responds within about one to two seconds of immersing the probes, and the alarm triggers correctly when either limit is crossed. Readings are stable once the averaging filter is applied. Accuracy depends strongly on calibration quality, probe condition and water temperature.

10. Advantages, Limitations, Applications and Cost

10.1 Advantages

- Low cost compared with commercial multi-parameter meters.
- Instant on-site results with a clear safe or unsafe decision.
- Both visual and audible alerts, useful in noisy or unattended places.
- Compact, portable and powered from a standard 5 V supply.
- Open design that is easy to modify, repair and extend.

10.2 Limitations

- Detects only pH and TDS; it cannot find bacteria, heavy metals, nitrates or pesticides.
- TDS does not identify which substances are dissolved.
- The pH probe needs regular calibration and careful storage.
- Low-cost sensors drift and have limited accuracy compared with laboratory instruments.
- Not suitable for continuous long-term immersion without maintenance.

10.3 Applications

- Homes and apartments, to check tap, borewell and purifier output.
- Schools and laboratories, as a teaching and demonstration tool.
- Aquariums, hydroponics and small-scale farming.
- Water purifier and RO plant monitoring.
- Small community water supply screening.

10.4 Approximate Cost Estimate

Item	Qty	Approx. cost (INR)
Arduino Uno / Nano	1	400 - 600
pH sensor module with probe	1	1,200 - 2,000
TDS sensor module	1	400 - 700
16x2 I2C LCD	1	200 - 300
LEDs, resistors, buzzer	set	50 - 100
Breadboard, wires, enclosure	set	200 - 400
Total		**about 2,500 - 4,100**

Prices are indicative and vary by supplier and region.

11. Future Scope, Conclusion and References

11.1 Future Scope

- **Turbidity sensor** to detect suspended particles and cloudiness.
- **Waterproof temperature sensor (DS18B20)** for accurate automatic compensation.
- **IoT connectivity** using ESP32 or ESP8266 to send data to a cloud dashboard or mobile app.
- **Data logging** to an SD card with a real-time clock to study trends over time.
- **SMS, email or app alerts** when readings cross limits.
- **Automatic actions** such as closing a valve or switching off a pump on unsafe water.
- **Solar power** and a rugged waterproof enclosure for field deployment.

11.2 Conclusion

The Water Quality Monitoring System shows that a reliable first-level check of drinking water can be made with inexpensive, widely available parts. By measuring pH and TDS, comparing them with accepted limits and giving immediate LED, buzzer and LCD feedback, the device turns raw sensor data into a clear and useful decision. It meets the stated objectives of real-time measurement, easy interpretation and low cost.

The system is best used as a screening and awareness tool. Where readings indicate a problem, or where health is at stake, the water should be confirmed by a certified laboratory. With additional sensors and wireless features, the design can grow into a complete remote monitoring platform.

11.3 References

[1] World Health Organization, *Guidelines for Drinking-water Quality*, 4th edition incorporating the first addendum, WHO.

[2] Bureau of Indian Standards, *IS 10500: Drinking Water - Specification*.

[3] Arduino documentation, *analogRead(), tone() and Wire library references*, arduino.cc.

[4] LiquidCrystal_I2C library documentation for HD44780 displays with PCF8574 backpacks.

[5] Manufacturer datasheets and wiki pages for the analog pH and TDS sensor modules used.

[6] APHA, *Standard Methods for the Examination of Water and Wastewater*.

12. Hardware Prototype and Output Screenshots

This page shows the assembled prototype and its outputs. The photographs document the hardware connections, the live values displayed on the 16x2 LCD, and the data printed on the Arduino IDE serial monitor during testing.

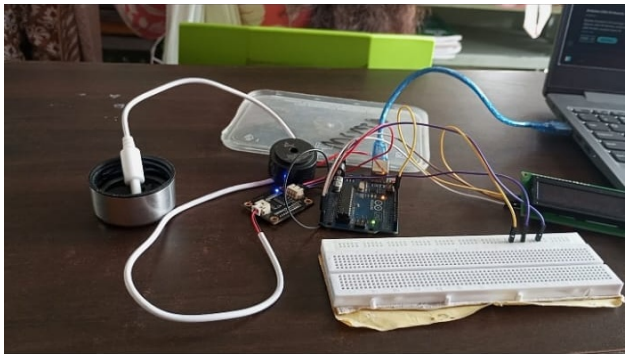


Figure 4: Prototype setup - Arduino Uno, sensor board, buzzer, breadboard and LCD connected to a laptop

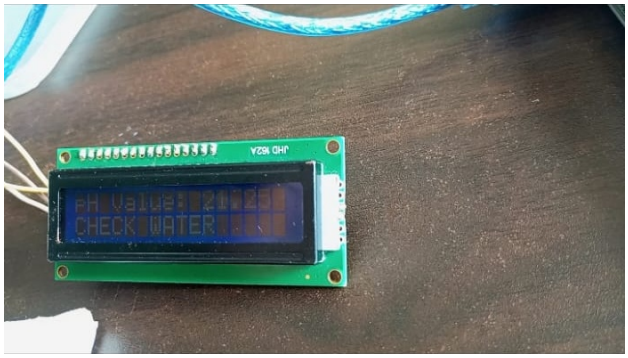


Figure 5: LCD showing the pH value with the "CHECK WATER" alert message

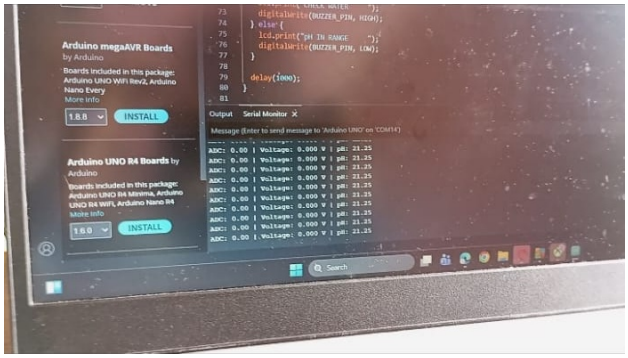


Figure 6: Arduino IDE serial monitor showing ADC, voltage and pH readings



Figure 7: LCD during start-up / refresh of the display

Observations: The LCD shows the pH value on the first line and the status message on the second line, and the alert message appears when the reading is outside the safe range. The serial monitor prints the raw ADC value, the converted voltage and the calculated pH on each cycle, which is useful for calibration and debugging.